AI Lab Test
2.7.2026

1. Draw the architecture of a **Multi-Layer Perceptron (MLP)** with **three hidden layers** containing **4, 3, and 2 neurons**, respectively, for classifying **4 × 4 grayscale images** into three classes: **dog, cat, and cow**.
2. Calculate the **total number of trainable parameters** (weights and biases) in your designed MLP.
3. How can you reduce the number of trainable parameters in the MLP you designed?
4. **Write a Python program** to implement the MLP architecture designed in the previous question. Your code should include the model definition, compilation, and a summary of the network architecture.
5. Write the format of the target labels (**y**) for your image classifier. Illustrate your answer by showing how the labels for the classes **dog**, **cat**, and **cow** would be represented.
6. Draw the architecture of a **Convolutional Neural Network (CNN)** with **three convolutional hidden layers**. Each convolutional layer should use **five 3 × 3 filters**. The network is designed to classify **4 × 4 grayscale images** into three classes: **dog, cat, and cow**.
7. For your CNN, clearly show the **number of feature maps** and the **spatial dimensions (height × width)** of the feature maps produced at each convolutional layer. State any assumptions you make regarding **padding** and **stride**.
8. **Write a Python program** to implement the CNN architecture designed in the previous question. Your code should include the model definition, compilation, and a summary of the network architecture.
9. Suppose you are given **45,000 labeled images** for training your image classifier. Determine how many images you would allocate to the **training**, **validation**, and **test** sets. Clearly justify your chosen data split based on machine learning best practices.
10. What **loss function(s)** would you use to train your CNN for this image classification task? Justify your choice based on the characteristics of the problem and the network's output.
11. Explain situations seeing curves shown in Fig.11(a) and Fig.11(b).

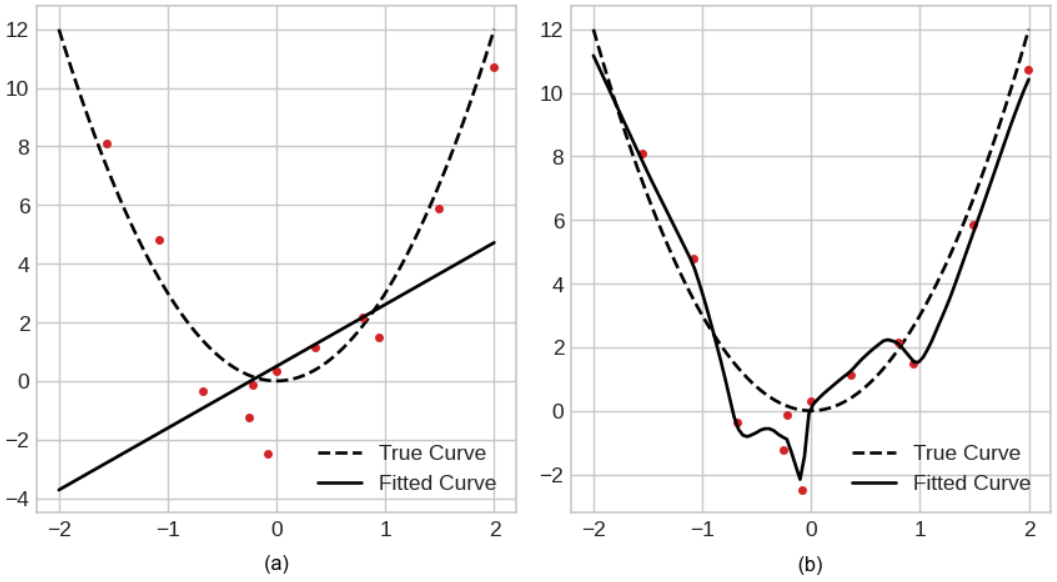


Fig.11